

78	$\left\ \begin{array}{l} \{75, 76, 77, 78\} \\ \{76, 77, 78, 79\} \\ \{77, 78, 79, 80\} \end{array} \right\ $	✓	The student who gets 75 will reply yes.
79			Another possibility is $S = \{34, 35, 36, 37\}$.
80			The student who gets 80 will reply yes.

Finally, all possibilities for S are $\{34, 35, 36, 37\}$, $\{47, 48, 49, 50\}$, $\{76, 77, 78, 79\}$, and $\{89, 90, 91, 92\}$, from which the answer is $37 + 50 + 79 + 92 = \boxed{258}$.

Remarks

- Alternatively, we can reconstruct the second table in this solution as follows, where Y and N denote the replies of "yes" and "no", respectively. Notice that this table has some kind of symmetry!

43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	
				Y	N	N	N			Y	N	N	N				Y	Y	Y	N						Y	N	N	N			Y	N	N	N							
					N	N	N	N				N	N	N	Y					N	Y	Y	Y					N	N	N	Y				N	N	N	N				

- As a confirmation, we can verify that each student will be able to deduce what S is upon hearing the four replies of "no" in unison. For example, if $S = \{47, 48, 49, 50\}$, then all students will know that no one gets 46 or 51, otherwise that student will reply yes (as discussed). Therefore, all students will conclude that S has only one possibility.

~MRENTHUSIASM

Solution 2

We know right away that $42 \notin S$ and $84 \notin S$ as stated in Solution 1.

To get a feel for the problem, let's write out some possible values of S based on the teacher's remarks. The first multiple of 7 that is two-digit is 14. The closest multiple of six from 14 is 12, and therefore there are two possible sets of four consecutive integers containing 12 and 14; $\{11, 12, 13, 14\}$ and $\{12, 13, 14, 15\}$. Here we get our first crucial idea; that if the multiples of 6 and 7 differ by two, there will be 2 possible sets of S without any student input. Similarly, if they differ by 3, there will be only 1 possible set, and if they differ by 1, 3 possible sets.

Now we read the student input. Each student says they can't figure out what S is just based on the teacher's information, which means each student has to have a number that would be in 2 or 3 of the possible sets (This is based off of the first line of student input). However, now that each student knows that all of them have numbers that fit into more than one possible set, this means that S cannot have two possible sets because otherwise, when shifting from one set to the other, one of the end numbers would not be in the shifted set, but we know each number has to fall in two or more possible sets. For example, take $\{11, 12, 13, 14\}$ and $\{12, 13, 14, 15\}$. The numbers at the end, 11 and 15, only fall in one set, but each number must fall in at least two sets. This means that there must be three possible sets of S , in which case the actual S would be the middle S . Take for example $\{33, 34, 35, 36\}$, $\{34, 35, 36, 37\}$, and $\{35, 36, 37, 38\}$. 37 and 34 fall in two sets while 35 and 36 fall in all three sets, so the condition is met. Now, this means that the multiple of 6 and 7 must differ by 1. Since 42 means the difference is 0, when you add/subtract 6 and 7, you will obtain the desired difference of 1, and similarly adding/subtracting 6 or 7 from 84 will also obtain the difference of 1. Thus there are four possible sets of S ; $\{34, 35, 36, 37\}$, $\{47, 48, 49, 50\}$, $\{76, 77, 78, 79\}$ and $\{89, 90, 91, 92\}$. Therefore the sum of the greatest elements of the possible sets S is $37 + 50 + 79 + 92 = \boxed{258}$.

~KingRavi

Solution 3

In a solution that satisfies these constraints, the multiple of 6 must be adjacent to multiple of 7. The other two numbers must be on either side.

WLOG assume the set is $\{a, 6j, 7k, b\}$. The student with numbers $a, 6j$, and $7k$ can think the set is $\{a - 1, a, 6j, 7k\}$ or $\{a, 6j, 7k, b\}$, and the students with number $6j, 7k$, and b can think the set is $\{a, 6j, 7k, b\}$ or $\{6j, 7k, b, b + 1\}$. Therefore, none of the students know the set for sure.

Playing around with the arrangement of the multiple of 6 and multiple of 7 shows that this is the only configuration viewed as ambiguous to all the students. (Therefore when they hear nobody else knows either, they can find out it is this configuration)

Considering S as $\{a, 6j, 7k, b\}$, b is $2 \pmod 6$ and $1 \pmod 7$, so b is $8 \pmod{42}$. (since it is all 2-digit, the values are either 50 or 92).

Similarly, considering S as $\{a, 7j, 6k, b\}$, b is $1 \pmod 6$ and $2 \pmod 7$, so b is $37 \pmod{42}$. The values that satisfy that are 37 and 79.

The total sum of all these values is therefore $50 + 92 + 37 + 79 = \boxed{258}$.

This solution will even work when the bounds (in this question, 2-digit so <100) are much larger and it is impractical to perform casework.

~balav123

Solution 4 (Logic)

Consider the tuple $(a, a + 1, a + 2, a + 3)$ as a possible S . If one of the values in S is 3 or $4 \pmod 7$, observe the student will be able to deduce S with no additional information. This is because, if a value is $b = 3 \pmod 7$ and S contains a $0 \pmod 7$, then the values of S must be $(b - 3, b - 2, b - 1, b)$. Similarly, if we are given a $b \equiv 4 \pmod 7$ and we know that $0 \pmod 7$ is in S , S must be $(b, b + 1, b + 2, b + 3)$. Hence, the only possibility for a is $5, 6 \pmod 7$.

In either case, we are guaranteed there is a $6, 0, 1 \pmod 7$ value in S . The difference comes down to if there is a $5 \pmod 7$ value or a $1 \pmod 7$ value. The person receiving such value will be able to determine all of S but the $6, 0, 1 \pmod 7$ people will not be able to differentiate the two cases ... yet.

Now consider which value among the consecutive integers is $c \equiv 3 \pmod 6$, if any. The person will know that S is either $(c - 3, c - 2, c - 1, c)$ or $(c, c + 1, c + 2, c + 3)$ to have a $0 \pmod 6$ value in S . Neither the $6, 1 \pmod 7$ person can be $3 \pmod 7$, else they can decipher what S is right off the bat by considering which set has $0 \pmod 7$. This translates to the possible $5 \pmod 7$ or $1 \pmod 7$ person cannot be $0 \pmod 6$. We are given that $0 \pmod 7$ and $0 \pmod 6$ cannot be the same person. Hence we conclude one of the $6 \pmod 7$ or $1 \pmod 7$ must be the $0 \pmod 6$ person.

Let the $6 \pmod 7$ person be $0 \pmod 6$. Then—hypothetical— $c \equiv 2 \pmod 7$ person is $3 \pmod 6$. After the first round, the $6, 0, 1 \pmod 7$ people realize that $2 \pmod 7$ is not in S else they would have deduced S by noting S was either $(c - 3, c - 2, c - 1, c)$ or $(c, c + 1, c + 2, c + 3)$ to have a $0 \pmod 6$ and choosing the former based on where $0 \pmod 7$ is. Hence they figure out S by knowing $5 \pmod 7$. So $a \equiv 5 \pmod 7$ and $a \equiv 5 \pmod 6$ (from $6 \pmod 7$ person being $0 \pmod 6$).

Similarly, if $1 \pmod 7$ person is $0 \pmod 6$ we find that $a \equiv 6 \pmod 7$ (so a $2 \pmod 7$ is in S) and $a \equiv 4 \pmod 6$.

By CRT, the possibilities are $a \equiv 5, 34 \pmod{42}$. The sum of the greatest values of S are the sum of $a + 3$ and so we get $(34 + 3) + (47 + 3) + (76 + 3) + (89 + 3) = \boxed{258}$.

~Aaryabhata1

Solution 5 (THINK)

Start by writing down every two-digit multiple of 6 and 7. First, 42 and 84 will not be in S because we need two distinct numbers where one is a multiple of 6 and ANOTHER is a multiple of 7. Furthermore, it is clear at this point that if the multiples of 6 and 7 are exactly 3 apart, then it will be obvious what S is; it is just the smaller multiple to the larger multiple. Therefore, we cross out numbers 18, 21, 24, 60, 63, 66.

That leaves us with pairs of multiples of 6 and 7 that are exactly 1 or 2 apart. In this case, if the multiples of 6 and 7 are two apart, such as 12 and 14, then the non-multiples will be able to deduce what the set is; this leaves that the 6 and 7 must be one apart.

Specifically, these are pairs (35, 36), (48, 49), (77, 78), and (90, 91). In each of these pairs, respectively, the largest number will be one greater than the larger multiple, namely 37, 50, 79, 92. Therefore, our answer is

$$37 + 50 + 79 + 92 = \boxed{258}.$$

~ xHypotenuse

Video Solution

<https://www.youtube.com/watch?v=7jKjilTRhs4>

Animated Solution by Interstigation

<https://youtu.be/TKXU33jnXkl>

~Interstigation

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